

WASP-12b

R-0314

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_240101 · created 2026-07-17T23:44:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460309.86861 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1242 +/- 0.0098
Transit depth (Rp/Rs)^2	1.54 +/- 0.24 [%]
Orbital Inclination (inc)	80.28 +/- 2.14
Airmass coefficient 1 (a1)	1.136 +/- 0.011
Airmass coefficient 2 (a2)	-0.0942 +/- 0.0066
Scatter in the residuals of the lightcurve fit is	0.93 %
Best Comparison Star	None
Optimal Aperture	4.43
Optimal Annulus	11.08
Transit Duration (day)	0.1174 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1242 $\pm$ 0.0098 vs 0.1178 (deviation 0.65 σ)
Duration fit vs published	0.1174 d vs 0.125 d (deviation 1.17 σ)
Mid-time O-C	-1575.3 min
Depth SNR	12.5
Residual scatter	0.93 %

Light curve

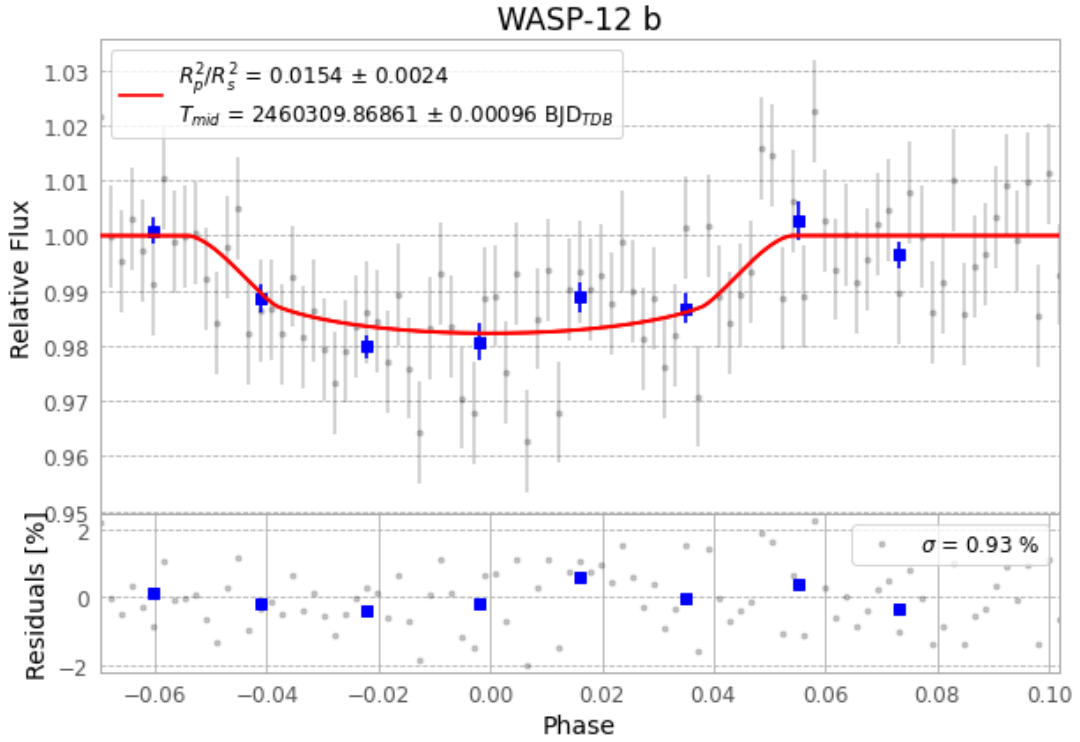


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-30.png (SHA-256 79e6df84be30b512...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 264], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 666e325fd36ba662...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

91 FITS frames (60 MB), WASP-12231231065115.FITS → WASP-12231231112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240101.log (ecd65ecd8417...), FinalLightCurve_WASP-12 b_2023-12-30.png (79e6df84be30...), FinalLightCurve_WASP-12 b_2023-12-30.csv (313fb72a5dd9...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 23:44Z.